

Biology, 1996 EC, entrance exam, booklet 43

Grade 12 · Biology · 1996 E.C.

Time allowed: 120 minutes

Questions: 49

Total marks: 49

Name _____ Grade _____ ID No. _____ Date _____

1. Select the type of behavior which is innate. (1)

A. Reflex action	B. Habituation
C. Conditioning	D. Imprinting

2. Select the field of biology that is concerned with the study of cells. (1)

A. Histology	B. Morphology
C. Cytology	D. Ecology

3. Which ecological interaction between two species results in a negative effect on both organisms? (1)

A. Mutualism	B. Competition
C. Predation	D. Neutralism

4. Which cellular structures are responsible for autolysis or self-digestion? (1)

A. Chromosomes	B. Ribosomes
C. Lysosomes	D. Golgi apparatus

5. What is the energy currency of the cell? (1)

A. DNA	B. Glucose
C. Starch	D. ATP

6. At which stage of cell division do homologous chromosomes line up at the equatorial plane of the cell? (1)

A. Prophase	B. Metaphase
C. Anaphase	D. Telophase

7. Which one of the following unicellular organisms is prokaryote? (1)

A. Amoeba	B. Bacteria
C. Euglena	D. paramecium

8. Which of the following is a postzygotic reproductive isolating mechanism? (1)

A. Seasonal isolation	B. Hybrid sterility
C. Mechanical isolation	D. Geographic isolation

9. Which of the following cells is diploid? (1)

A. Egg	B. sperm
C. Second polar body	D. Zygote

10. To which one of the following groups of organic molecules do enzymes belong? (1)

A. Carbohydrates	B. Lipids
C. Proteins	D. Nucleic acids

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- 11.** Choose the animal group that is represented in Ethiopia by a higher proportion of endemic species in relation to the total number of species of that animal group found in the country. (1)
- A. Birds
B. Reptiles
C. Amphibians
D. Fishes
-
- 12.** Which term applies to the rearing of fish under more or less controlled conditions? (1)
- A. Apiculture
B. Pisciculture
C. Silviculture
D. Aquaculture
-
- 13.** Which of the following is the middle layer of an embryo? (1)
- A. Neural plate
B. Ectoderm
C. Mesoderm
D. Endoderm
-
- 14.** Based on our mode of nutrition, to which of the following groups do we humans belong? (1)
- A. Ominvores
B. Carnivores
C. Herbivores
D. Autotrophs
-
- 15.** What is the best term used to refer to the association of different species of organisms interacting and living together? (1)
- A. Ecosystem
B. Community
C. Pyramid
D. Niche
-
- 16.** Choose the one that represents anabolic reaction (1)
- A. Deamination
B. Photosynthesis
C. Digestion
D. Aerobic respiration
-
- 17.** What is a plasmid? (1)
- A. Material that is found in blood
B. Material found in the cell membrane
C. The same as a virus
D. A strand of DNA in bacteria cell
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- 18.** Which type of dominance operates when it is known that both alleles of a gene contribute to the features of the phenotype? (1)
- A. Complete dominance
B. Incomplete dominance
C. Codominance
D. Intermediate dominance
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- 19.** If tissues from sliced potato tuber are kept in sea water which of the following is more likely to happen? (1)
- A. The cells will take up water and burst
B. The cells lose water and become flaccid
C. The cells become dry and impermeable.
D. The cells actively pump out salts
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- 20.** To which of the following properties of life does the term irritability refer? (1)
- A. Excretion
B. Response to stimulus
C. Homeostasis
D. Metabolism
-
- 21.** From which of the following do scientists usually formulate a hypothesis? (1)
- A. Observation
B. Prediction
C. Theory
D. Experiment

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- 22.** Select the alternative that contains only vector borne infections. (1)
- A. Malaria, Trachoma, Syphilis, Tapeworm
 - B. Malaria, Leshmaniasis, Trypanosomiasis, Typhus
 - C. Diphtheria, Tuberculosis, Cholera, Meningitis
 - D. Leprosy, Ringworm, Pneumonia, Hookworm
-
- 23.** Which of the following groups probably ate more of a herbivorous diet? (1)
- A. Homo erectus
 - B. Homo habilis
 - C. Cromagnon humans
 - D. Australopithecus afarensis
-
- 24.** Which new research technique was used to produce the new plant now known as the "pomato"? (1)
- A. Genetic engineering
 - B. Protoplast fusion
 - C. Sexual hybridization
 - D. Inbreeding
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- 25.** To which tropic level do green plants belong? (1)
- A. Zero
 - B. First
 - C. Second
 - D. Third
-
- 26.** If an F_1 round yellow pea of $R_r Y_y$ genotype is selfed, which class of pea is the most frequent in the F_2 generation? (1)
- A. Wrinkled green
 - B. Round green
 - C. Wrinkled yellow
 - D. Round yellow
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- 27.** Which property of water keeps lake-inhabiting organisms from freezing in winter in cold regions like the Arctic region? (1)
- A. High latent heat of vaporization
 - B. High heat of fusion
 - C. High surface tension
 - D. Low density of the solid state
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- 28.** Which of the following enzymes has its optimum pH within low pH range? (1)
- A. Pepsin
 - B. Salivary amylase
 - C. Pancreatic lipase
 - D. Catalase
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- 29.** Suppose an archeologist discovered remains of plant material which contains one-eighth of its original content of carbon-14. If the half-life of carbon-14 is 5600 years, how old is the plant material? (1)
- A. 50,000 years
 - B. 28,000 years
 - C. 16,800 years
 - D. 11,200 years
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- 30.** Which of the following happens during the dark reaction of photosynthesis? (1)
- A. ATP is produced
 - B. NADPH is produced
 - C. Oxygen is split from water
 - D. Carbon dioxide is fixed
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- 31.** Which enzyme is used in textile industry to remove starch used in fabrics as adhesive or support during weaving? (1)
- A. Amyloglucosidase
 - B. Lipase
 - C. Amylase
 - D. Trypsin

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- 32.** Which of the following crop species is NOT an oil crop? (1)
- | | |
|--------------------------------------|-----------------------------------|
| A. <i>Guizotia abyssinica</i> | B. <i>Arachis hypogaea</i> |
| C. <i>Lens culinaris</i> | D. <i>Sesamum indicum</i> |
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- 33.** Which two groups of organisms does the fossil animal known as *Archeopteryx* link? (1)
- | | |
|-----------------------------------|---------------------------------|
| A. Reptiles and birds | B. Mammals and reptiles |
| C. Reptiles and amphibians | D. Amphibians and fishes |
-
- 34.** During protein synthesis, which of the following molecules transports amino acids to their correct positions in the growing chain of polypeptide? (1)
- | | |
|-------------------------|---------------------------------|
| A. Messenger RNA | B. Deoxyribonucleic acid |
| C. Ribosomal RNA | D. Transfer RNA |
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- 35.** If the protoplasm shrinks away from the cell wall when a plant cell is bathed in a sugar solution, what is the concentration of the solution relative to the protoplasm of the plant cell? (1)
- | | |
|----------------------|----------------------|
| A. Hypertonic | B. Isotonic |
| C. Hypotonic | D. Isoosmotic |
-
- 36.** What is the method of conservation in which animals are kept in captivity and plants are grown in botanical gardens? (1)
- A.** In-situ conservation
B. Zoo conservation
C. Ex-situ conservation
D. Complementary in-situ and ex-situ conservation
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- 37.** In which region of Ethiopia would you look for an ecosystem dominated by lowland dry forests? (1)
- | | |
|---------------------------|---------------------------|
| A. Afar region | B. Somali region |
| C. Gambella region | D. Southern region |
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- 38.** A chimpanzee concentrates to reach at a bunch of grapes hanging above its head. What kind of behavior is this? (1)
- | | |
|--------------------------------|----------------------------------|
| A. Insight learning | B. Latent learning |
| C. Associative learning | D. Sensitization learning |
-
- 39.** Which factor is least responsible for the reduction or loss of biodiversity in Ethiopia? (1)
- | | |
|--|------------------------------|
| A. Over-harvesting biological resources | B. Climate change |
| C. Human cultural diversity | D. Commercial farming |
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- 40.** In the life cycle of a moss plant, which of the following is diploid? (1)
- | | |
|----------------------------------|---------------------------------|
| A. Gametophyte generation | B. Sporophyte generation |
| C. Protonema | D. Spores |
-
- 41.** If a strand of DNA carrying AAT bases is copied to RNA what would the corresponding bases be in the RNA? (1)
- | | |
|---------------|---------------|
| A. UUA | B. TTA |
| C. AAT | D. UUT |
-
- 42.** If the letter “d” is observed under a microscope, what would its image look-like? (1)
- | | |
|---------------------------|--------------------|
| A. Letter d itself | B. Letter b |
| C. Letter q | D. Letter p |

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- 43.** Which of the following best illustrates loss of biodiversity? (1)
- A. The death of an individual insect that belongs to a successful species
 - B. A complex forest ecosystem is converted to a field of a modern high yielding crop variety
 - C. A new plant species is introduced from a neighboring country and becomes part of the flora
 - D. A species develops a number of distinct varieties
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- 44.** Choose the alternative that best illustrates the recombinant DNA technique. (1)
- A. Artificial insemination of animals
 - B. Crossing related plants to improve yield.
 - C. Gene from one organism are introduced into the genome of an unrelated organism
 - D. Crossing of carefully selected varieties of a species
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- 45.** Which of the following diseases of animals would be controlled if one is able to control tsetse flies? (1)
- A. Heartwater
 - B. Rinderpest
 - C. Trypanosomiasis
 - D. Blackleg
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- 46.** Suppose a couple produced a legitimate child of blood type B, what are the genotypes of the couple? (1)
- A. AA × BB
 - B. AA × AB
 - C. AA × BO
 - D. AO × BO
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- 47.** Which of the following techniques is employed in the transfer of human insulin gene to bacteria in order to produce human insulin in the bacteria? (1)
- A. DNA finger printing technique
 - B. Protoplast fusion technique
 - C. Cross pollination technique
 - D. Recombinant DNA technique
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- 48.** Suppose a man has a dominant gene on his X-chromosome, Which of his children would show the trait that is controlled by this gene? (1)
- A. All sons and daughter
 - B. All sons only
 - C. All daughter only
 - D. Half of the sons and half of the daughters
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- 49.** In Ethiopia, which of the following genera of microorganisms has the potential to be used for single cell protein production? (1)
- A. Spirulina
 - B. Bacillus
 - C. Rhizobium
 - D. Agrobacterium

— END OF EXAMINATION —